

Exercise 1

Schema

```
Emp(eid:integer, ename:string, age:integer, salary:real)
Works(eid:integer, did:integer, pct_time:integer)
Dept(did:integer, dname:string, budget:real, managerid:integer)
```

Note

- Works.*pct_time* is the employee percent working time. It is expressed as a number between 0 (not working) and 100 (full-time).

Queries

1. Print the names and ages of each employee who works in both the Hardware department and the Software department.
2. For each department with more than 20 full-time-equivalent employees (i.e., where the part-time and full-time employees add up to at least that many full-time employees), print the *did* together with the number of employees that work in that department.
3. Print the name of each employee whose salary exceeds the budget of all of the departments that he or she works in.
4. Find the *managerids* of managers who manage only departments with budgets greater than \$1 million.
5. Find the *enames* of managers who manage the departments with the largest budgets.
6. If a manager manages more than one department, he or she *controls* the sum of all the budgets for those departments. Find the *managerids* of managers who control more than \$5 million.
7. Find the *managerids* of managers who control the largest amounts.
8. Find the *enames* of managers who manage only departments with budgets larger than \$1 million, but at least one department with budget less than \$5 million.

Solutions exercise 1

1.

```
SELECT E.ename, E.age
FROM Emp E, Works W1, Works W2, Dept D1, Dept D2
WHERE E.eid=W1.eid AND W1.did=D1.did AND D1.dname='Hardware'
AND E.eid=W2.eid AND W2.did=D2.did AND D2.dname='Software'
```
2.

```
SELECT did, COUNT(eid)
FROM Works
GROUP BY did
HAVING 2000 < SUM(pct_time);
```
3.

```
SELECT ename
FROM Emp E
WHERE salary>ALL(
  SELECT budget
  FROM Dept D, Works W
  WHERE E.eid = W.eid AND D.did = W.did
);
```
4.

```
SELECT managerid
FROM Dept
GROUP BY mangerid
HAVING MIN(budget) > 1000000;

-- Alternative
SELECT DISTINCT D.managerid
FROM Dept D
WHERE 1000000 <ALL(
  SELECT D2.budget
  FROM Dept D2
  WHERE D2.managerid=D.managerid
);
```
5.

```
SELECT ename
FROM Emp
WHERE eid IN (
  SELECT managerid
  FROM Dept
  WHERE budget=(
    SELECT MAX(budget)
    FROM Dept
  )
);
```
6.

```
SELECT managerid
FROM Dept
GROUP BY mangerid
HAVING SUM(budget) > 5000000;

-- Alternative
SELECT D.managerid FROM Dept D
WHERE 5000000 < (
  SELECT SUM(D2.budget)
  FROM Dept D2
  WHERE D2.managerid=D.managerid
);
```

7. `SELECT DISTINCT` managerid
`FROM` (
 `SELECT DISTINCT` managerid , `SUM`(budget) `AS` tot_budget
 `FROM` Dept
 `GROUP BY` managerid
)
`AS` control
`WHERE` control.tot_budget=(
 `SELECT MAX`(tot_budget)
 `FROM` control
);
8. `SELECT` ename
`FROM` Emp E
`JOIN` Dept D `ON` eid = managerid
`GROUP BY` eid , ename
`HAVING` `EVERY`(budget>1000000)
 `AND ANY`(budget<5000000)